Exercise 1: SQL Fundamentals

1. SELECT Statement

```
SELECT *
FROM employees;
```

id	first_name	last_name	department	salary	hire_date	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

2. SELECT DISTINCT Statement

```
SELECT DISTINCT department
FROM employees;
```

department
IT
HR
Finance
Marketing

3. ORDER BY Statement

```
SELECT first_name,
       last_name,
       salary
FROM employees
ORDER BY salary DESC;
```

first_name	last_name	Salary
Emily	Davis	62000
Mike	Johnson	60000
Robert	Wilson	59000
John	Doe	55000
Sarah	Brown	53000
Daniel	Clark	53000
David	White	52000
Jessica	Moore	51000
Laura	Hall	50000
Jane	Smith	48000

4. LIMIT Statement

```
SELECT id,
       salary
FROM employees
ORDER BY salary DESC
LIMIT 5;
```

id	salary
6	62000
3	60000
7	59000
1	55000
4	53000

5. WHERE Statement

```
SELECT id,
       department
FROM employees
WHERE department = 'IT';
```

id	department
1	IT
4	IT
6	IT
10	IT

6. AND Statement

```
SELET id,
       department,
       salary
FROM employees
WHERE department = 'Finance' AND salary > 58000;
```

id	department	salary
3	Finance	60000
7	Finance	59000

7. OR Statement

```
SELECT id,
       department
FROM employees
WHERE department = 'HR' OR department = 'Marketing';
```

id	department
2	HR
5	Marketing
8	HR
9	Marketing

8. NOT Statement

```
SELECT id,
       department
FROM employees
WHERE NOT department = 'IT';
```

id	department
2	HR
3	FINANCE
5	Marketing
7	Finance
8	HR
9	Marketing

9. IN Statement

```
SELECT id,
       department
FROM employees
WHERE department IN ('HR', 'IT', 'Finance');
```

id	department
1	IT
2	HR
3	Finance
4	IT
6	IT
7	Finance
8	HR
10	IT

10. Combining Conditions

```
SELECT id,
    department,
    salary
FROM employees
WHERE department = 'IT' AND salary > 50000 AND city = 'New York';
```

id	department	salary
1	IT	55000
4	IT	53000

11. Combining WHERE, AND, and ORDER BY

```
SELECT first_name,
       last_name,
       department,
       salary
FROM employees
WHERE department = 'Finance' OR department = 'Marketing' AND salary > 52000
ORDER BY salary DESC;
```

first_name	last_name	department	salary
Mike	Johnson	Finance	60000
Robert	Wilson	Finance	59000
Daniel	Clark	Marketing	53000

12. Combining SELECT DISTINCT, WHERE, and IN

```
SELECT DISTINCT city,
                id,
                department
FROM employees
WHERE department IN ('Finance', 'Marketing');
```

city	id	department
Los Angeles	3	Finance
San Francisco	5	Marketing
Houston	7	Finance
Chicago	9	Marketing

13. Combining WHERE, NOT, AND, and ORDER BY

```
SELECT id,
       department,
       salary,
       hire_date
FROM employees
WHERE NOT department='Finance' AND salary >50000
ORDER BY hire_date ASC;
```

id	department	salary	hire_date
6	IT	62000	2015-02-14
5	Marketing	52000	2016-04-10
8	HR	51000	2018-05-22
1	IT	55000	2018-06-15
4	IT	53000	2021-03-25
9	Marketing	53000	2022-06-01

14. Combining WHERE, OR, IN, and LIMIT

```
SELECT id,
       city,
       department
FROM employees
WHERE city IN('Chicago', 'Los Angeles') AND department IN('IT', 'Marketing')
LIMIT 3;
```

id	city	department
6	Chicago	IT
9	Chicago	Marketing